

Eventless Payload-Adaptive NMPC for Quadrotors

Anonymous Authors

Affiliation withheld for double-anonymous review

Abstract—A release command does not guarantee that a payload has physically detached, so a controller that clears its payload model on that command can fly a loaded vehicle as if it were empty. We present an eventless payload-model interface for quadrotor nonlinear model predictive control (NMPC) that never treats attach or release commands as evidence. A moving horizon estimator (MHE) [1] estimates composite mass and the horizontal first mass moment $s = m_{Pr_{xy}}$ from odometry and a rotor-speed thrust proxy that is not computed from the payload estimate; unlike a center-of-mass offset, s remains well defined as the payload vanishes. The inertia increment follows from the mass estimate and a weak vertical-arm prior, and health-gated, rate-limited updates drive both the NMPC model and body-rate scheduling. In PX4 [2] SITL with physically attached eccentric payloads, including a 0.30 kg, 4 m/s stress test at the vehicle’s torque limit, release confirmation completed in 32 of 36 flights in which the payload attached and the vehicle survived to the release command; all four failures, at 4 m/s, trace to estimator failure. Confirmation took a median 3.46 s in maneuver and 16.56 s after braking to hover. When detachment lagged the command by 4 s, a baseline clearing on the command did so too early in 5/5 flights, but persistent confirmation, with or without the command, never did; when no command reached the controller, the interface still detected unsigned payload loss in 10 of 18 flights. These results are limited to simulation and the tested conditions.

I. INTRODUCTION

Autonomous aerial delivery and in-flight aerial manipulation subject a multirotor to abrupt changes in its own dynamics. Grasping or releasing a parcel changes three coupled quantities at once: the total mass m (which rescales the thrust-to-acceleration map and the hover throttle), the center of mass (a thrust that no longer passes through the composite CoM produces a constant trim torque $\tau \approx \Delta r \times f$, significant even for small offsets because $f \approx mg$ is large), and the rotational inertia (which slows the attitude response and effectively lowers the inner-loop bandwidth). The first channel is an outer-loop (position/thrust) problem; the latter two are inner-loop (attitude/rate) problems. A controller that models the payload only as a scalar mass will mistrack or, for sufficiently heavy eccentric loads, destabilize.

Explicit parameter estimation (EKF/MHE of mass, CoM, inertia) is interpretable and can feed model-based constraints, but converges slowly across an abrupt change; *lumped-disturbance adaptation* (L1-adaptive NMPC, INDI) reacts fast but yields no physical model; *learning-based* estimators typically need rich sensing or ground-truth data. We take the explicit route deliberately and accept its latency: the proposed interface trades release-confirmation delay for semantics that never run ahead of physical evidence and for named physical parameters.

The central question of this work is whether the complete grasp–carry–release lifecycle can adapt its payload model without trusting a commanded or externally supplied payload event. Our answer rests on two observations. First, payload mass and first moment can be estimated continuously, with the inertia increment following algebraically from the mass estimate and a weak geometric prior, without consuming a gripper command, attachment state, or external event notification. Second, rotor-speed telemetry provides a collective-thrust and body-torque proxy that is *not computed from* the online payload estimate, under a calibrated rotor thrust model; residuals formed from it therefore do not inherit a mass error through the controller’s own command. Combining the two, the estimator needs no external event signal and no measurement of the horizontal attachment geometry.

Contributions.

- 1) A **first-moment payload interface**. Estimating mass together with the horizontal first mass moment—the standard linear parameterization of rigid-body inertial parameters [3]—keeps the payload geometry well defined through release, where a CoM offset is not, and lets one health-gated $m/s/J$ frame drive bounded NMPC model updates and continuous body-rate scheduling (§III–IV).
- 2) **Release semantics that do not run ahead of evidence**: the controller clears its payload model only after the estimator’s no-payload confidence persists, never on its own release command, and enters UNRESOLVED and brakes to hover when evidence is missing (§IV).

We evaluate the complete interface in PX4 SITL, reporting every attempt, the confirmation path each flight took, a paired delayed-detachment fault injection, unsigned partial payload loss, and the failures observed at the vehicle’s actuator limit (§VI-A, §VII). The runtime estimator consumes only motor speeds and odometry; the sole geometric quantity not identified online is a weak vertical-arm prior r_z , which is structurally unobservable from the thrust-moment channel.

II. RELATED WORK

Agile quadrotor flight. Geometric tracking on SE(3) [4] and time-optimal flight [5] assume known, fixed inertial parameters, which a payload event invalidates mid-flight.

Aerial manipulation and transport. Aerial robots that grasp, carry and release objects are surveyed in [6]; dynamic aerial grasping, with explicitly modeled gripper dynamics, was demonstrated in [7]. Load transportation, by grasping or cable suspension, is surveyed in [8]. Ours is the transport-and-release subproblem: the payload is rigidly attached, so the controller

must handle the resulting change in mass, inertia and centre of mass.

Control under changing payload. Adaptive and robust designs—L1-adaptive NMPC [9], INDI [10]—treat the load as a lumped disturbance and are the standard baselines for payload-carrying flight; L1-NMPC in particular reports large tracking-error reductions under unknown load on aggressive trajectories. A parallel line learns the residual dynamics instead: [11] fits unmodeled aerodynamic effects with Gaussian processes inside the MPC, which improves tracking without yielding a named physical parameter either. Slung-load control addresses the swinging-payload case explicitly [12], exploiting differential flatness of the cable-suspended system; a rigidly attached payload instead changes the vehicle’s own inertial parameters. A related proprioceptive line estimates the external wrench directly with momentum-based observers, enabling collision detection and interaction control [13]. All of these recover a *force/torque*, not a physical parameter: they cannot, e.g., update a thrust-margin constraint or an inner-loop gain schedule from the estimate.

MHE and online parameter estimation. Filter- and optimization-based estimators recover mass, inertia, and geometric parameters [14]–[16], and the mass and CoM of a grasped payload have been estimated for aerial manipulation [17]. Writing the CoM as a first mass moment makes inertial parameters enter the dynamics linearly, a standard device in manipulator load identification [3]; we use it because it remains well defined when the payload vanishes. Standard MHE performance is highly sensitive to the process/measurement weight matrices; neural MHE (NeuroMHE) [18] learns to output these weights online, resolving the bias–variance trade-off but estimating a lumped disturbance force rather than explicit named parameters.

Residual-based change detection. Residual-based fault/change detection (CUSUM and related methods [19]) motivates the evidence checks used by the payload lifecycle state machine. Our positioning: we do not learn dynamics and add no sensing; every deployed signal derives from motor speeds and odometry. This is a deliberate departure from learning-based MPC [20] and the broader safe-learning agenda [21], where a learned component absorbs the model mismatch; here the mismatch is resolved into named parameters that downstream constraints can consume.

III. SYSTEM OVERVIEW AND PROBLEM SETUP

Architecture (Fig. 1). Three layers: Gazebo physics (with a magnetic DetachableJoint gripper producing genuine inertia/CoM changes), the PX4 inner-loop rate controller, and a ROS 2 outer loop (NMPC + MHE). The outer loop publishes a collective thrust and a body-rate setpoint; PX4 tracks the body rate.

Rigid-body model. Both optimizers integrate the same 13-state rigid-body model. Stacking position p , velocity v , attitude quaternion q and body rate ω into $x = [p^\top, v^\top, q^\top, \omega^\top]^\top \in$

$\mathbb{R}^{13}$, with input $u = [T, \tau^\top]^\top$ (collective thrust and body torque),

$$\begin{aligned}\dot{p} &= v, & \dot{q} &= \frac{1}{2} q \otimes [0, \omega^\top]^\top, \\ \dot{v} &= \frac{1}{m} (R(q) e_3 T - k_d v) - g e_3, \\ \dot{\omega} &= J^{-1} (\tau + \tau_{\text{com}} - \omega \times J \omega),\end{aligned}\tag{1}$$

where $\otimes$ is the Hamilton product, $R(q)$ the body-to-world rotation, e_3 the vertical unit vector (ENU, z up), and k_d a linear drag coefficient. The first line is kinematics and is independent of the payload; the payload enters the second and third lines only through the inertial parameters. Those are the composite mass m , the horizontal first mass moment s , and the inertia J . Thrust acts at the geometric center, so a composite CoM displaced by c_{xy} produces a trim torque, and the payload adds a parallel-axis inertia increment:

$$\begin{aligned}c_{xy} &= \frac{s}{m}, & \mu &= \frac{m_B \max\{m - m_B, 0\}}{m}, \\ \tau_{\text{com}} &= T [-c_y, c_x, 0]^\top = \frac{T}{m} [-s_y, s_x, 0]^\top, \\ J &= \text{diag}(J_{B,xx} + \Delta J, J_{B,yy} + \Delta J, J_{B,zz}), \\ \Delta J &= \mu r_z^2,\end{aligned}\tag{2}$$

where m_B is the known bare-airframe mass and r_z a weak vertical-arm prior from the payload rack. The small $O(r_{xy}^2)$ diagonal terms and the yaw increment are omitted in the deployed interface. Equations (1)–(2) are what the NMPC integrates; the MHE integrates the same equations with one addition given below. The two optimizers differ only in how (m, s) reach them: the NMPC receives them as parameters refreshed before every solve, the MHE carries them as states.

Predictive controller. The NMPC model carries the runtime parameter vector $p = [x_{\text{ref}}, m, \Delta J, c_{xy}]$, i.e. the reference plus the inertial parameters of (2). The NMPC solves

$$\begin{aligned}\min_{u_{0:N-1}} & \sum_{k=0}^{N-1} (\|e_k\|_Q^2 + \|u_k - \bar{u}(p)\|_R^2) + \|e_N\|_{Q_N}^2 \\ \text{s.t. } & x_{k+1} = F(x_k, u_k, p), \quad x_0 = \hat{x}, \quad u_k \in \mathcal{U},\end{aligned}\tag{3}$$

where x and u are those of (1), e_k stacks the position, velocity, quaternion-vector and rate errors to the reference, and F is one RK4 step of (1) ($\Delta t = 0.05$ s, $N = 20$). The input reference $\bar{u}(p) = [mg, c_y mg, -c_x mg, 0]^\top$ is the live hover input including the eccentric trim torque. Q and R follow Bryson’s rule with tolerances 0.1 m, 0.3 m/s, 0.05, 0.3 rad/s, 2 N and the torque bounds; Q_N scales the position, attitude and rate blocks of Q by 20, 2 and 0.2. $\mathcal{U}$ is the actuator envelope $T \in [1.27, 31.35]$ N, $|\tau_x|, |\tau_y| \leq 0.5$ N m, $|\tau_z| \leq 0.2$ N m. The deployed MHE augments both composite mass and the horizontal first mass moment,

$$\begin{aligned}x_{\text{MHE}} &= [x^\top \quad m \quad s_x \quad s_y]^\top \in \mathbb{R}^{16}, \\ s &= m_P r_{xy},\end{aligned}\tag{4}$$

with $\dot{m} = 0$ and $\dot{s} = 0$ within each window ($N = 20$, $\Delta t = 0.1$ s).

What s is and where it acts. Let m_B be the known bare-airframe mass and $m_P = (m - m_B)^+$ the payload mass. The augmented s is the horizontal *first mass moment*

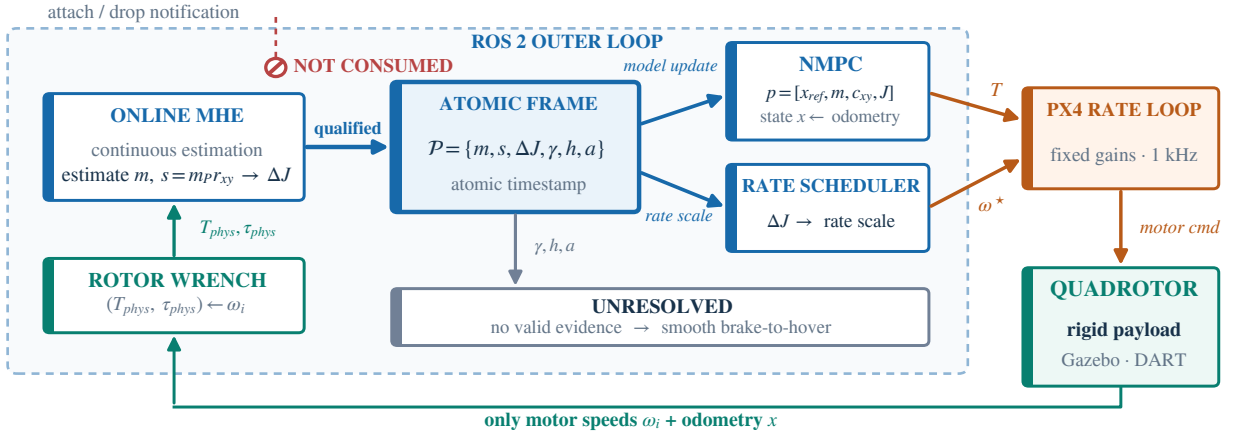


Fig. 1. System architecture and information flow. The full pipeline consumes only motor speeds and odometry; one atomic, health-gated $m/s/J$ frame drives both the NMPC model and continuous rate-command scheduling. No attach/drop notification enters this path.

of the composite body about the body-frame origin. The bare airframe's CoM coincides with that origin, so the airframe contributes nothing and the entire moment is the payload's, $s = mPr_{xy}$, for whatever grasp offset r_{xy} the gripper happens to realize. This is the standard linear parameterization of rigid-body inertial parameters [3]: the rotational dynamics contains the product mPr_{xy} , never the two factors separately, which is why the product—not the offset—is the state we augment. The two augmented states then enter F_M —the same (1) driven by u^{phys} —through different channels. The MHE is the one place where the off-diagonal inertia is retained,

$$J_{xz} = -\frac{m_B}{m}s_x r_z, \quad J_{yz} = -\frac{m_B}{m}s_y r_z, \quad (5)$$

added to the J of (2). So s acts in exactly two places: the trim torque τ_{com} , and (5). It appears nowhere in the translational equation, which carries m alone. Mass is therefore identified predominantly through the translational residual and the moment exclusively through the angular-rate residual; both are components of the same odometry measurement y_j , so the second state costs no additional sensing. Neither quantity is measured: both are recovered as the window-constant values that reconcile the reconstructed input u^{phys} with the observed v and ω trajectories. The geometry of (2) is not a second estimator but a reparameterization of this same state for the controller, whose model keeps its 13 states and receives m , c_{xy} and ΔJ as the parameters p of (3). The MHE solves

$$\begin{aligned} \min_{x,w} & \|x_0 - \bar{x}_0\|_{W_0}^2 \\ & + \sum_{j=0}^{N-1} \beta_j (\|y_j - Cx_j\|_{W_y}^2 + \|w_j\|_{W_w}^2) \\ \text{s.t. } & x_{j+1} = F_M(x_j, u_j^{\text{phys}}) + Gw_j, \\ & m \in [0.95 m_B, 5 \text{ kg}], \quad |s_x|, |s_y| \leq 1.5 \text{ kg m}, \end{aligned} \quad (6)$$

over $x_{0:N}$ and $w_{0:N-1}$, where y_j is the odometry and C selects the 13 vehicle states, $u^{\text{phys}} = [T_{\text{phys}}, \tau_{\text{phys}}]^T$ is reconstructed from rotor speeds (below), and $G = [I_{13}, 0]^T$ keeps m and s noise-free. W_y is the inverse squared odometry noise (0.02 m,

0.05 m/s, 0.01, 0.02 rad/s); $W_w = \text{diag}(10^4 I_6, 10^5 I_4, 10^3 I_3)$ deliberately trusts the dynamics. The arrival cost $W_0 = \text{diag}(10^3 I_6, 10^4 I_4, 10^2 I_3, 0.1, 100 I_2)$ is centred on the previous window's shifted solution and is nearly uninformative in m ($\sigma \approx 3.2 \text{ kg}$) and s ($\sigma = 0.1 \text{ kg m}$). In the reported 4 m/s conditions $\beta_j \equiv 1$. An earlier residual-triggered schedule, which scaled stages preceding a self-detected T_{phys} step (4.4 N from a slow baseline) by 10^{-4} , remained armed in the 2 m/s flights but never fired there; it was disabled at 4 m/s because grasp and release transients crossed its threshold. No external event enters β_j . The geometry published to the controller is reconstructed from the solved (m, s) by (2).

Estimate-independent thrust proxy. Each rotor produces $F_i = k\omega_i^2$ with $k = 8.55 \times 10^{-6} \text{ N s}^2$, and the full wrench follows from the rotor positions (x_i, y_i) and spin directions $d_i \in \{\pm 1\}$,

$$T_{\text{phys}} = \sum_i F_i, \quad \tau_{\text{phys}} = \begin{bmatrix} \sum_i y_i F_i \\ -\sum_i x_i F_i \\ -k_m \sum_i d_i F_i \end{bmatrix}, \quad (7)$$

with $|x_i| = |y_i| = 0.174 \text{ m}$ and yaw-reaction coefficient $k_m = 0.016 \text{ m}$. This is the *actuator-generated* wrench: it excludes $J\dot{\omega}$ and rotor gyroscopic terms, which is what makes it usable as the known input u^{phys} of (6) rather than as a second estimate of the state. Rotor speeds arrive at 250 Hz and are averaged over each 10 Hz MHE interval, the zero-order-hold-consistent discretization of (1). The quadratic coefficient k is a static-thrust constant; it is known to lose validity under large induced inflow during fast translation [22], which we do not model. Feeding the NMPC's *commanded* thrust back into the estimator would instead close an estimator–controller loop: that command is computed from $\hat{m}$ itself (its cost anchors T near $\hat{m}g$), and any inversion error, saturation, or motor lag between command and rotor is invisible to the estimator, so a mass error can persist self-consistently rather than appear as a residual. The rotor-speed proxy is taken downstream of those stages and is not computed from $\hat{m}$; it does depend on the

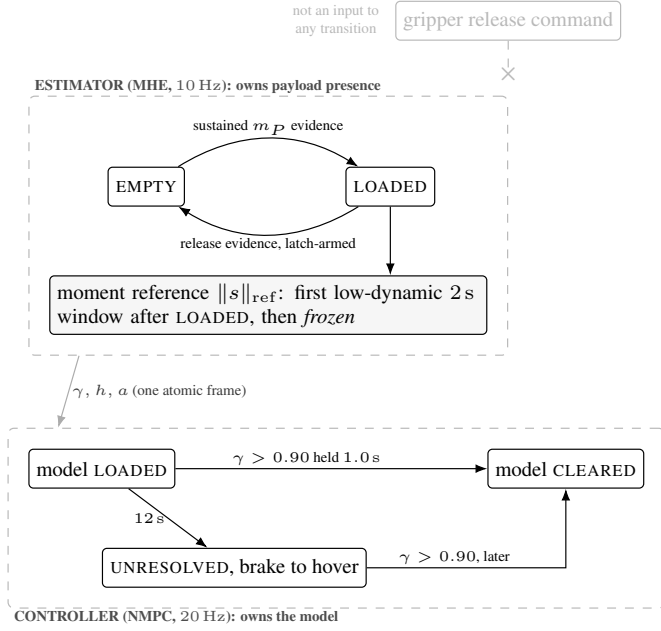


Fig. 2. The payload lifecycle as two state machines. The estimator owns payload *presence* and may declare release from the evidence of (10); the controller owns the *model* and clears it only on confirmation, $\gamma > 0.90$ held for 1.0 s. The gripper release command enters neither machine. An unhealthy or stale frame freezes the last accepted model and resets the persistence timer, so a degraded estimate can delay a confirmation but never cause one. UNRESOLVED is a hold, not a clearing: the model stays loaded while the vehicle brakes to hover and keeps waiting for evidence.

calibrated k , and in our simulator the plant generates thrust with the same square law (§VII).

Thrust-to-throttle inversion. The offboard collective-thrust command is inverted to a normalized rotor command through the motor model $T = k\omega^2$. Provided that the forward thrust reconstruction and inverse command mapping use the same k , steady-state closed-loop tracking is generally not directly affected by its absolute value.

Continuous inner-loop scheduling. Model consistency in the NMPC cannot by itself restore the PX4 rate loop's lost bandwidth. The deployed path therefore maps the live inertia increment to a body-rate-command scale, low-pass filters it, applies target hysteresis and asymmetric rise/recovery slew limits, and preserves 5% command headroom before the final saturation. It does *not* rewrite PX4 parameters. An attach-time inertia-only bootstrap uses the rack's 0.30 kg maximum payload for immediate safety, then fades out after persistent coherent MHE evidence; neither payload mass nor s is injected by this bootstrap, and the drop path remains estimate-only.

IV. ONLINE ESTIMATION AND EVENTLESS ADAPTATION

A. Continuous payload interface and autonomous lifecycle

The MHE publishes one atomic frame

$$\mathcal{P} = \{m, s_x, s_y, c_x, c_y, \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz}, J_{xx}, J_{yy}, J_{zz}, \gamma, h, a\}, \quad (8)$$

where γ is no-payload confidence, h is a health bit and a is solution age. Health requires a successful latest solve and fresh odometry, motor-speed, and known-input streams. The NMPC additionally checks local receive age; an unhealthy or stale frame freezes the last accepted model and breaks the confidence persistence timer. Valid frames pass through a low-pass filter and separate slew limits for m , s , and ΔJ before every NMPC solve.

Payload presence is an estimator-owned state, distinct from any externally signalled attach state. A hysteretic EMPTY $\leftrightarrow$ LOADED machine uses sustained payload-mass evidence, with a directional residual as a fast path, and a separate latch records that the present flight reliably entered LOADED. This latch prevents an empty-flight confidence value from consuming the later drop confirmation and prevents a transient mass dip during a figure-eight from disarming the true release path.

Decision variables. Every lifecycle decision is a function of the published frame (8) alone. Writing $\sigma(v; a, b)$ for the smooth indicator that is 1 for $|v| \leq a$, 0 for $|v| \geq b$, and $1 - 3x^2 + 2x^3$ with $x = (|v| - a)/(b - a)$ in between, the no-payload confidence fuses three empty indications multiplicatively,

$$\gamma = \sigma(m_P; 0.015, 0.060) \sigma(\|s\|; 0.0015, 0.0060) \times \sigma(\Delta J_{xy}; 0.0020, 0.0100), \quad (9)$$

with thresholds in kg, kg m and kg m² and $\Delta J_{xy} = \max(|\Delta J_{xx}|, |\Delta J_{yy}|)$. The product is deliberate: any single payload indication vetoes EMPTY, so a nonzero moment keeps γ low even when $\hat{m}$ touches its lower bound. Release evidence uses two binary indications,

quantity-empty : $m_P < 0.030$ kg,

moment-collapse : $\rho = \frac{\|s\|}{\|s\|_{\text{ref}}} < 0.30 \wedge \|s\| < 0.008$ kg m. (10)

The absolute bound is a sanity limit, not a discriminator: over five development flights the loaded and released $\|s\|$ distributions overlapped (loaded minimum 0.0039 kg m below released maximum 0.0049 kg m), so only the ratio separates them.

The loaded reference. ρ presumes a reference the estimator must form by itself. After the LOADED declaration, a healthy frame is admitted to a candidate window only if $\|s\| \leq 0.039$ kg m—the platform envelope $m_{P,\max} r_{xy,\max}$, beyond which a sample is physically impossible, discarded, and counted as an estimator-quality diagnostic—and the vehicle is low-dynamic, $\|\omega\| \leq 0.15$ rad/s and $\|v_{xy}\| \leq 0.20$ m/s; any maneuver clears the window. Once 20 consecutive frames (2 s) accumulate, they must further be stable in magnitude ($1.4826 \text{ MAD/median} \leq 0.25$) and consistent in direction ($\|\frac{1}{20} \sum_j s_j / \|s_j\|\| \geq 0.90$, i.e. the mean unit moment direction over the window is near unit length). The reference is then the 20%-trimmed mean of the window magnitudes, and is *frozen* for the remainder of the flight: a running maximum would absorb the figure-eight estimation errors it is meant to detect. If no window qualifies before the aggressive segment,

ρ never becomes available and the moment-gated path cannot fire—the dominant miss mechanism in §VII.

On accepted release evidence, the estimator switches only the *published target* of s to zero; the optimization state is not overwritten. The output filter consequently produces a continuous decay rather than a geometric step. The controller marks a planned drop complete, resets its adaptive state, and finishes recovery only after healthy confidence remains above 0.90 for 1.0 s; issuing the gripper command itself does none of these things.

B. Why the first mass moment is the essential state

Estimating CoM offset alone is ill-conditioned at release: physically $s = m_P r_{xy} \rightarrow 0$, whereas r_{xy} itself is undefined as $m_P \rightarrow 0$. In an earlier mass-only formulation, the optimizer could suppress a phantom eccentricity only by driving total mass below the bare-airframe value; 14 of 55 post-release samples then pinned at the lower bound. Augmenting s supplies the rotational channel directly and permits $c_{xy} = s/m$ without knowing the grasp-dependent horizontal offset. A frozen-window inertia coefficient prevents the optimizer from using instantaneous mass as a free inertia knob, while the weak r_z prior retains the dominant parallel-axis term. It is the smallest structure we found that represents payload existence, trim moment, and roll/pitch inertia without an external attachment geometry measurement.

Estimator declaration and controller confirmation. The controller clears its payload model through one route only: confirmation, i.e. $\gamma > 0.90$ held for 1.0 s on healthy, fresh frames. The estimator may additionally *declare* release, which switches the published target of s to zero and so speeds the rise of γ ; a declaration is neither necessary nor sufficient for confirmation. The final code has two declaration routes. (a) A *moment-gated detector* fires on the two indications of (10) jointly for five frames (slow path), or on moment collapse accompanied by a persistent signed thrust-residual step $\Delta T = \bar{T}_{\text{post}} - \bar{T}_{\text{pre}}$ over 0.3 s half-windows, below -0.9 N and significant against a rolling robust σ of past ΔT , for four frames (moment-residual path). Moment collapse is mandatory in both. (b) A *mass-domain route* declares release on the quantity-empty indication alone, held for 20 frames and accumulated only in low-dynamic flight. Confirmation can also occur without any declaration; the moment factor of (9) still vetoes it while $\|s\|$ is nonzero. Neither (b) nor confidence alone uses the frozen-reference ratio, so confirmations through them are not evidence for (a); Sec. VI-A reports how often each occurred. Moment evidence must come from a fresh, successful solve: re-anchoring after five consecutive solver failures initializes s to zero, and that value is not counted as collapse. During development we also tested a quantity-residual branch (a 0.9 N residual plus quantity-empty evidence, without moment evidence); it was removed after premature releases (Sec. VI-A). A recorded figure-eight transient reached -1.69 N, larger in magnitude than the -1.47 N step of the smallest payload, so residual amplitude alone cannot separate the cases. Online and offline evaluation call the same pure

TABLE I
LIFECYCLE THRESHOLDS AND THEIR PROVENANCE. “EARLIER FLIGHTS”
PREDATE, AND ARE DISJOINT FROM, THE REPORTED CAMPAIGN.

threshold	value	provenance
moment-collapse ratio ρ	0.30	leave-one-out cross-validation on 59 earlier flights; every fold chose 0.30 (0.20 succeeded in 48/59)
residual floor / persistence	0.9 N; 2 or 4 frames	below the 1.47 N weight of the smallest payload; on a recorded flight, figure-eight excursions above 0.9 N lasted ≤ 3 frames, the release step 4 frames
slow-path persistence	5 frames (0.5 s)	engineering choice
mass-domain release	$m_P < 0.030$ kg, 20 frames	between the loaded minimum (0.079 kg) and post-release maximum (0.007 kg) of earlier 0.15 kg flights
no-payload confidence	0.90 for 1.0 s	engineering choice
UNRESOLVED timeout / brake	12 s / 3 s	engineering choice

decision function, and an UNRESOLVED timeout is treated as unknown, never as proof of an empty vehicle.

Table I lists the lifecycle thresholds and where each value comes from. All were fixed in code before the final campaign began and were not retuned on it. Only ρ and the residual floor were selected against data from earlier flights; the remaining values are engineering choices whose closed-loop sensitivity we have not swept.

V. SIMULATION SETUP

PX4 SITL with Gazebo [23] (Harmonic), an x500 quadrotor, and a magnetic DetachableJoint gripper that produces genuine inertia and CoM changes.

The physical DetachableJoint gripper attaches a separate rigid body and therefore couples mass, inertia and CoM together. It supports the lifecycle, baseline, and safety claims in the paper; no wrench-emulated mass-step result is retained.

Task sequence (Fig. 3): approach (position control) $\rightarrow$ NMPC hand-off and warmup hover $\rightarrow$ controlled descent $\rightarrow$ active grasp enable $\rightarrow$ lift $\rightarrow$ figure-eight $\rightarrow$ phase-triggered release. We state the NMPC scope explicitly: it governs the grasp hover and the post-event adaptation phase; the approach controller is not part of the claim. Batches run headless, with a stand-in process supplying the ground-station heartbeat that PX4 requires.

Real-time budget. Both optimizers are acados [24] SQP solvers with merit backtracking and the HPIPM [25] QP solver. On one desktop core shared with Gazebo/PX4 SITL, the deployed 13-state NMPC ($N = 20$, 20 Hz) solves in 1.3 ms at the median and 2.2 ms at the 99th percentile ($n = 1009$), i.e. 2.6 % and 4.4 % of its 50 ms period. The deployed 16-state first-moment MHE ($N = 20$, 10 Hz) solves in 9.2 ms at the median, 24.6 ms at the 90th and 59.2 ms at the 99th percentile, with a 72.8 ms maximum ($n = 3740$, first 20 frames per flight discarded), against a 100 ms period—roughly $7\times$ the 1.4 ms median of the historical mass-only estimator. This meets the deadline in simulation, but the tail margin is narrow, so we make *no* onboard-headroom claim; that requires measurement

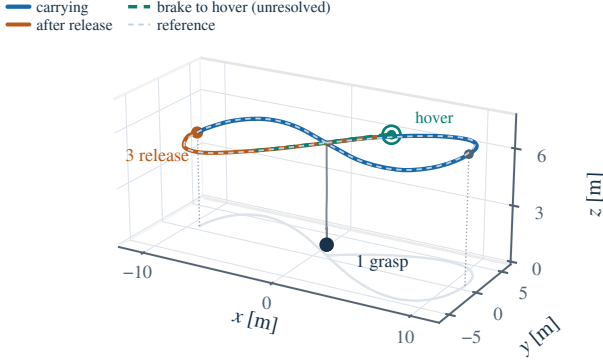


Fig. 3. Measured mission trajectory (simulator ground truth), two 0.15 kg/2 m/s flights of the same batch: grasp (1), lift, then carry along the figure-eight with ± 0.8 m vertical undulation (blue) against the nominal reference (pale dashed). After the release command (3) one flight clears the moment vote in maneuver and keeps tracking (orange); the other declares UNRESOLVED at its 12 s timeout and brakes to a hover (teal), confirming at 16.40 s. The two paths agree to within 0.12 m (median) before the timeout; loaded tracking RMS is 0.053 m, and 0.060 m after release.

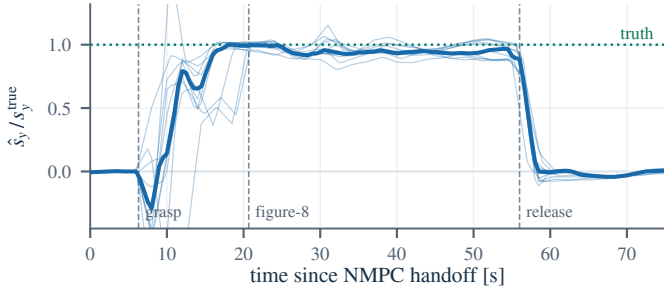


Fig. 4. First mass moment against ground truth (0.15 kg at 2 m/s, $n = 11$, each normalised by its own truth). Truth is reconstructed offline as $m_P r_y$ from the simulator-measured grasp offset and never reaches the estimator, which runs with no horizontal attachment geometry. $\hat{s}_y$ is zero before grasp, rises with no attach notification, holds near truth through the figure-eight, and decays to zero after release. The carry-phase estimate is biased toward zero, as the batch shown here illustrates; the cause is not identified. The moment-gated detector uses the ratio $\|s\|/\|s\|_{\text{ref}}$, which is insensitive to a common scale bias.

on the target computer. Setpoints are published at 50 Hz by interpolating the latest prediction.

VI. RESULTS

Provenance. The lifecycle cohort uses the final 16-state first-moment interface; its two 4 m/s conditions were re-flown with the residual-triggered schedule disabled, and all lifecycle flights predate the window-based re-initialization seed of §VII.

A. Eventless grasp–release lifecycle

The controller issues the gripper-release command itself but does not use it to clear its payload model. We report every eventless flight in which the payload physically attached (from the gripper’s attach record) and the vehicle survived to the release command. The pre-registered rule judged attachment from MHE mass evidence instead, which silently excluded flights whose estimator failed to recognize an attached payload.

TABLE II

PAIRED COMMAND-TRIGGERED ABLATION. ARM A CLEARS THE MODEL AT THE COMMAND BY CONSTRUCTION; ARM B REQUIRES ESTIMATOR EVIDENCE; ARM C IS ARM B WITH ITS RELEASE EVIDENCE ARMED ONLY BY THE COMMAND. THE LAST TWO ROWS INJECT A COMMAND/PHYSICAL-DETACHMENT MISMATCH; “EARLY” MEANS MODEL CLEARING BEFORE THE DETACHABLEJOINT IS REMOVED (“–”: NOT SCORED). THE ONE VALID UNPAIRED ARM B RUN OF THE A/B HOLD ROW ALSO DID NOT CLEAR EARLY.

condition	pairs	A compl.	B compl.	early clear	A/B
0.15 kg, 4 m/s	8	8/8	8/8	–	
0.30 kg, 4 m/s	3	3/3	3/3	–	
0.15 kg, 2 m/s	8	8/8	8/8	–	
nominal pooled	19	19/19	19/19	–	
0.15 kg, 2 m/s + 4 s hold	5	5/5	5/5	5/5 / 0/5	
same, arm C instead of A	6	6/6 (C)	6/6	0/6 / 0/6	

a) Paired command-triggered baseline: We retain the pre-registered paired design rather than selecting arm B alone after the comparison. Arm A is a command-triggered baseline on the legacy interface: it treats the release command as the detachment event and immediately resets payload geometry and inner-loop scaling. Arm B is the proposed interface. The arms also differ before release: arm A uses the legacy, non-continuous geometry path and schedules the inner loop by rewriting PX4 rate gains, whereas arm B uses the continuous $m/s/J$ frame and body-rate-command scaling. A pair enters the comparison only when both arms are eligible. Across 19 such pairs (8/3/8 across the three conditions) both arms complete 19/19 releases with no observed post-release instability (Table II); arm A has zero confirmation delay by definition. Only three 0.30 kg/4 m/s pairs formed because arm A crashed during or immediately after lift in 11 of 16 attempts at that condition (Sec. VII).

Under delayed detachment a command-triggered baseline is expected to clear prematurely; the experiment measures this temporal mismatch and the confirmation delay that evidence-based updates incur. A fault injector received the normal release command but retained the physical DetachableJoint for a median 4.01 s. Of twelve arm-runs, one arm A run never armed PX4, leaving five valid pairs and one valid unpaired arm B run. In all five pairs arm A cleared its payload model at the command, a median 4.01 s before physical detachment. Arm B cleared before detachment in none of its six valid runs: it entered UNRESOLVED in 5/6 and then confirmed a median 12.45 s after detachment, and the remaining run confirmed 3.00 s after detachment. All eleven valid runs had an independent Gazebo DETACHED record. Median position-error peaks during the hold were 0.071 m (A) and 0.068 m (B), so the result supports a difference in model–plant consistency, not a tracking advantage or crash avoidance. We did not test a permanently jammed gripper, under which arm A would carry a loaded vehicle with an empty model.

b) Command-armed confirmation and unsignaled loss:

The hold result reflects persistent confirmation, not eventlessness: a variant armed only by the command (arm C, otherwise identical to arm B, confirming on a 3 s thrust drop) also

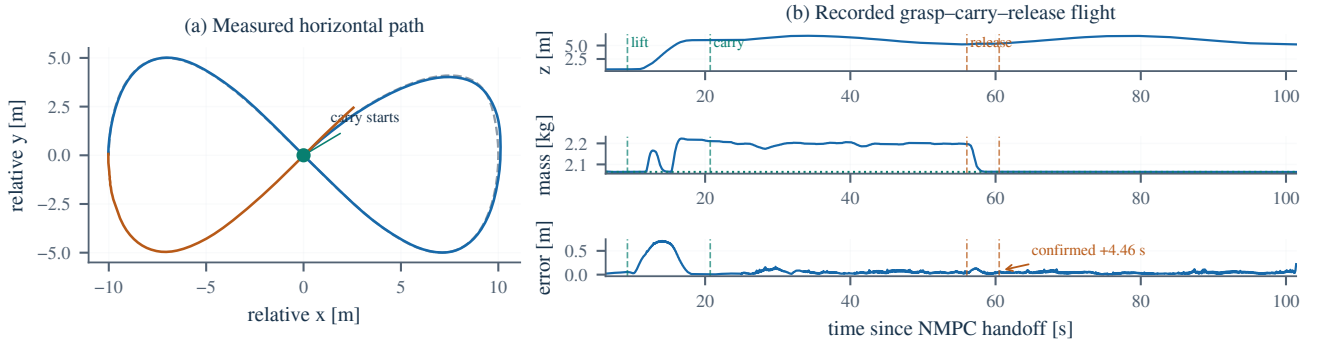


Fig. 5. One eventless flight from the final batch (0.15 kg, 2 m/s). Left: the measured horizontal path (blue while carrying, orange after release) against the nominal reference including its amplitude ramp (grey dashed). Right: recorded height, controller-consumed mass estimate, and position-error norm through grasp, lift, carrying, release command, and confirmation. This flight confirms in maneuver, 4.46 s after the command; it illustrates one execution and does not replace the cohort statistics in Table III.

never cleared early (0/6 pairs) and confirmed sooner after detachment (median 4.3 s versus 7.3 s). Eventlessness matters when no release command reaches the controller—a gripper that opens by itself, a load torn off, or a release triggered elsewhere whose notification is lost. With such losses injected at the release phase, arm B detected the loss in 10 of 18 paired flights (4/9 at 2 m/s, 4/5 and 2/4 at 0.15 kg and 0.30 kg/4 m/s; median latency 2.2 s to 2.9 s), while arm C, never armed, detected none and kept a phantom first moment. Seven of the eight misses occurred because the loaded moment reference had not formed, or the payload had not been recognized, before the loss. The cost, a detector exposed throughout loaded flight, was not observed: no release was declared before a command or injected loss in 209 loaded flights (3.7 h; per-flight 95 % upper bound 1.4 %).

c) Completion and delay of the complete interface:

The complete interface confirmed release in 32 of 36 eligible flights (Table III; two-sided 95 % Clopper–Pearson interval [73.9 %, 96.9 %]). All four failures, at 4 m/s, trace to estimator failure (Sec. VII): three never recognized the payload and stayed UNRESOLVED, and one diverged after release. The pre-registered rule would have excluded the first three (31/32). This is a conditional closed-loop result for the interface as a whole, not a validation of any single release detector: Table IV shows that three confirmation paths occurred.

The confirmation delay is *bimodal*. In 17 of the 32 completed flights the confidence crosses threshold while the vehicle is still flying the figure-eight, and confirmation completes in a median 3.46 s (Fig. 5). In the remaining 15 no release evidence is available in maneuver— $\hat{s}$ has drifted under sustained excitation or the frozen reference never formed—so the system reaches its 12 s timeout, declares UNRESOLVED, and executes a 3 s smooth brake to hover; confirmation then completes at a median 16.56 s. Table III keeps the two modes separate.

No flight in any path declared release before the command. In 23 of the 32 completed flights the payload-presence machine made exactly two transitions, one at grasp and one at release; in the nine confidence-only flights it never declared EMPTY within the logged window (over 40 s past confirma-

TABLE III
RELEASE COMPLETION AND CONFIRMATION DELAY (ALL ELIGIBLE ARM B FLIGHTS). DELAY RUNS FROM THE RELEASE COMMAND TO THE END OF THE 1.0 s CONFIDENCE HOLD AND IS COMPUTED OVER COMPLETED FLIGHTS. THE TWO DELAY COLUMNS ARE DISTINCT CONTROL PATHS AND ARE NOT POOLED.

condition	n	compl.	delay median [IQR]	
			in-maneuver	via brake-to-hover
0.15 kg, 4 m/s	11	10/11	2.90 s [2.90, 2.90]	16.60 s [16.60, 16.60]
0.30 kg, 4 m/s	13	10/13	8.95 s [8.00, 9.26]	16.40 s [16.20, 16.70]
0.15 kg, 2 m/s	12	12/12	3.31 s [2.95, 4.36]	16.48 s [16.40, 16.50]
pooled	36	32/36	3.46 s [2.90, 8.00]	16.56 s [16.46, 16.60]

Overall delay max 16.70 s. Share confirming only after brake-to-hover: 50 %, 40 %, 50 % of completed flights (rows in order), 47 % pooled.

TABLE IV
CONFIRMATION PATHS IN THE 32 COMPLETED FLIGHTS OF TABLE III. EVERY CONFIRMATION USED THE CONTROLLER’S CONFIDENCE ROUTE (§IV); COLUMNS RECORD THE ESTIMATOR DECLARATION, IF ANY. *Moment-gated*: THE DETECTOR DECLARED BEFORE THE CONFIDENCE CROSSED. *Mass-domain*: THE FROZEN REFERENCE NEVER FORMED; THE CONFIDENCE CROSSED DURING OR SHORTLY AFTER THE BRAKE TO HOVER AND THE MASS-DOMAIN ROUTE DECLARED 0.9 s TO 3.1 s LATER. *Confidence only*: THE REFERENCE NEVER FORMED AND NO DECLARATION OCCURRED (ALL IN MANEUVER).

condition	n	moment-gated	mass-domain	confidence only
0.15 kg, 4 m/s	10	9	1	0
0.30 kg, 4 m/s	10	1	3	6
0.15 kg, 2 m/s	12	9	0	3
pooled	32	19	4	9

tion). Every load declaration came from sustained mass evidence; no external grasp signal was consumed. The published first moment decayed continuously rather than stepping to zero (Fig. 4).

d) *Partial payload loss*: Four 0.05 kg boxes were grasped together and three were detached one at a time (45, 80 and 115 s after grasp) with no signal and no release command; a control arm carried the same boxes without loss (five alternating valid pairs per speed). Pre-registered criteria: (a) $\hat{m}$ enters ± 0.02 kg of each new truth in ≥ 80 %

of steps; (b) paired tracking-RMS difference ≤ 0.02 m, no crash; (c) no EMPTY declaration after the first two losses. At 2 m/s all criteria pass: $\hat{m}$ followed 14 of 15 steps (median convergence 1.1 s to 8.8 s) with no false EMPTY and tracking indistinguishable from the control arm. At 4 m/s (a) reached only its threshold (12/15), so we claim recovery at 2 m/s only.

e) Development-stage evidence: Two earlier tests motivated the final design but used earlier code and do not validate the final detector. A quantity-residual fast branch without moment evidence declared release prematurely in 3/9 randomized flights during a 4 m/s figure-eight. After its removal, a 12-flight randomized test of the moment-gated detector, run before the frozen reference was introduced, produced no pre-command declaration, 11 moment-gated releases (median detector delay 1.9 s), and one flight that stayed UNRESOLVED.

f) Attempts and exclusions: The paired campaign comprised 84 attempts. Arm B (proposed) made 42: 36 eligible, five crashed before the release command (three in the 0.30 kg/4 m/s figure-eight, two during lift at 2 m/s), and one did not physically attach. Arm A made 42: 25 eligible and 17 pre-release crashes, 11 of them during or just after lift at 0.30 kg/4 m/s. Because the arms differ before release, these rates are reported per arm rather than pooled or compared, and neither characterizes end-to-end mission reliability. Table II uses the 19 pairs with both arms eligible; Table III uses all 36 eligible arm B flights and Table IV the 32 that completed. Tail-window tracking is not compared, because arm B brakes to hover after an unresolved timeout while arm A continues the figure-eight.

VII. LIMITATIONS AND FUTURE WORK

All system-level evidence is presently SITL-only. Hardware migration replaces simulated motor speeds with ESC telemetry and must re-identify thrust/torque coefficients, noise floors, timing, and gripper compliance before free flight.

Thrust proxy. In simulation the plant uses the same square law $k\omega^2$ as the estimator, and motor speeds are exact. Scaling only the estimator's thrust map by $\pm 5\%$ (0.15 kg, 2 m/s, four repetitions per arm, 11 valid flights) caused no pre-command release in any arm but defeated the interface in both directions. At 0.95 the payload was never recognized (0/3) and every flight stayed UNRESOLVED; at 1.05 a phantom 0.09 kg persisted after release (4/4), every flight stayed UNRESOLVED, and one diverged. The matched 1.00 arm recognized the payload in 4/4. Thus k must be calibrated well within 5%; speed noise, delay and induced inflow [22] remain untested.

Missing baselines. The only baselines reported here are the command-triggered and command-armed arms of §VI-A, which isolate release semantics. We have not run any comparison against a fixed-parameter NMPC, against a lumped-disturbance adaptation such as L1-adaptive NMPC [9] or INDI [10], or against a CUSUM detector [19] on the same signals under a matched false-alarm constraint. The vertical arm r_z is unobservable from the thrust-moment channel and remains a weak prior; the s_x /pitch channel is noisier than s_y /roll. The

deployed diagonal inertia approximation also drops the small $O(r_{xy}^2)$ and yaw-inertia terms.

Release-detector boundary. First-moment augmentation is now part of the default path. The development-stage tests and the three final-batch confirmation paths are reported in Sec. VI-A; 32/36 completions do not establish distribution-wide detector robustness. In 13 of the 32 completed flights the healthy, low-dynamic, in-envelope reference window never filled after the LOADED declaration, so the moment-ratio test never ran. Out-of-envelope $\hat{s}$ samples are excluded from moment votes but still reach the controller's c_{xy} through filtering and slew limits. The envelope assumed a 0.13 m grasp offset while the gripper admitted 0.15 m; this affected 2 of 13 0.30 kg flights and does not explain most missing references. Reliable reference formation therefore remains necessary for a robust autonomous release detector.

All reported release batches explicitly enable the release-evidence path. Its built-in default is off, under which every flight ends UNRESOLVED, so deployment requires an explicit enable plus startup validation of the configuration.

First-moment drift. Sustained figure-eight excitation can increase $\|\hat{s}\|$ from 0.012 kg m to 0.031 kg m; withholding moment evidence then delays confirmation. Deceleration restored the vote within 3.3 s to 3.4 s in the affected test flights. Together with missing-reference cases, this yields the brake-to-hover delays in Table III. An online counter of first-moment envelope exceedances correlates with this outcome at 4 m/s (in-maneuver confirmation 9/10 for zero exceedances versus 2/10 otherwise; Fisher $p = 0.006$), but misses all six delayed 2 m/s cases. Earlier exit from the maneuver remains unimplemented and requires validation.

Operating limit. 0.30 kg at 4 m/s is a stress test at the vehicle's actuator limit, not an operating point. Hover needs 74 % of maximum thrust, but torque binds: in the figure-eight, pitch torque saturated within 98 % of 5-s diagnostic windows and yaw within 53 % (46 %/16 % at 0.15 kg, $\leq 6\%$ at 2 m/s), and body-rate scaling ran at a median 4.39 of its 5.0 cap. Three of the five arm B pre-release crashes (figure-eight divergences) and three of the four incomplete releases occurred there, and the legacy arm A crashed during or just after lift in 11 of 16 attempts because its rate scaling starts from the lower-bound mass estimate while the box is still grounded. Every eligible 2 m/s flight confirmed release. We claim safe release semantics, not reliable flight, at this limit; deployment needs torque margin.

Estimator failures. In four flights the published mass never left the empty value (in two, repeated solver failures re-initialized from a lower-bound seed); the three at 4 m/s flew without diverging and stayed UNRESOLVED. In one 0.30 kg flight the estimate was unhealthy for 48 s before release; because the phase-timed release command is not gated on estimator health, the frozen model no longer matched the vehicle after detachment and it diverged. A window vertical-force balance now seeds re-initialization (pre-registered A/B at this condition: figure-eight re-initializations 29 to 1); the reported cohort predates it, and health-gated release remains

open.

A sustained vertical up/downdraft remains intrinsically confounded with a payload change on a collective-thrust-only residual; wind is not evaluated in the reported cohort.

Simulator fidelity. (i) gz-sim Harmonic cannot change a link's inertia at runtime in a way the physics adopts (issue #2733), so every payload change is produced by the DetachableJoint platform, which attaches a separate rigid body (§V). (ii) The payload is rigidly attached; gripper compliance and pendulum modes are not modeled. Neither affects which signals the estimator consumes.

VIII. CONCLUSION

We presented an eventless payload-model interface for quadrotor NMPC that estimates composite mass and the horizontal first mass moment from odometry and a rotor-speed thrust proxy, and that clears its payload model only on persistent estimator evidence, never on its own release command. In PX4 SITL the complete interface confirmed release in 32 of 36 eligible flights through three confirmation paths, with median delays of 3.46 s in maneuver and 16.56 s after braking to hover; all four failures arose from estimator failure at 4 m/s, three of them at the vehicle's torque limit. Under a 4 s delayed detachment, persistent confirmation, with or without the command, avoided the early model clearing of a command-triggered baseline (5/5); the eventless interface additionally detected 10 of 18 unsignaled payload losses that a command-armed confirmer cannot see. Hardware validation, thrust-map mismatch, reliable moment-reference formation, gating release on estimator health, and stronger baselines remain open.

REFERENCES

- [1] C. V. Rao, J. B. Rawlings, and D. Q. Mayne, "Constrained state estimation for nonlinear discrete-time systems: stability and moving horizon approximations," *IEEE Transactions on Automatic Control*, vol. 48, no. 2, pp. 246–258, 2003.
- [2] L. Meier, D. Honegger, and M. Pollefeys, "PX4: A node-based multithreaded open source robotics framework for deeply embedded platforms," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 6235–6240.
- [3] C. G. Atkeson, C. H. An, and J. M. Hollerbach, "Estimation of inertial parameters of manipulator loads and links," *The International Journal of Robotics Research*, vol. 5, no. 3, pp. 101–119, 1986.
- [4] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *IEEE Conference on Decision and Control (CDC)*, 2010, pp. 5420–5425.
- [5] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Science Robotics*, vol. 6, no. 56, p. eabh1221, 2021.
- [6] A. Ollero, M. Tognon, A. Suarez, D. Lee, and A. Franchi, "Past, present, and future of aerial robotic manipulators," *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 626–645, 2022.
- [7] J. Thomas, G. Loianno, J. Polin, K. Sreenath, and V. Kumar, "Toward autonomous avian-inspired grasping for micro aerial vehicles," *Bioinspiration & Biomimetics*, vol. 9, no. 2, p. 025010, 2014.
- [8] D. K. D. Villa, A. S. Brandão, and M. Sarcinelli-Filho, "A survey on load transportation using multirotor UAVs," *Journal of Intelligent & Robotic Systems*, vol. 98, no. 2, pp. 267–296, 2020.
- [9] D. Hanover, P. Foehn, S. Sun, E. Kaufmann, and D. Scaramuzza, "Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 690–697, 2022.
- [10] E. J. J. Smeur, Q. Chu, and G. C. H. E. de Croon, "Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles," *Journal of Guidance, Control, and Dynamics*, vol. 39, no. 3, pp. 450–461, 2016.
- [11] G. Torrente, E. Kaufmann, P. Foehn, and D. Scaramuzza, "Data-driven MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 3769–3776, 2021.
- [12] K. Sreenath, N. Michael, and V. Kumar, "Trajectory generation and control of a quadrotor with a cable-suspended load — a differentially-flat hybrid system," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 4888–4895.
- [13] T. Tomić, C. Ott, and S. Haddadin, "External wrench estimation, collision detection, and reflex reaction for flying robots," *IEEE Transactions on Robotics*, vol. 33, no. 6, pp. 1467–1482, 2017.
- [14] J. Svacha, J. Paulos, G. Loianno, and V. Kumar, "IMU-based inertia estimation for a quadrotor using Newton–Euler dynamics," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3861–3867, 2020.
- [15] V. Wüest, V. Kumar, and G. Loianno, "Online estimation of geometric and inertia parameters for multirotor aerial vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 1884–1890.
- [16] C. Böhm, M. Scheiber, and S. Weiss, "Filter-based online system-parameter estimation for multicopter UAVs," in *Robotics: Science and Systems (RSS)*, 2021, paper 087, DOI 10.15607/RSS.2021.XVII.087.
- [17] D. Mellinger, Q. Lindsey, M. Shomin, and V. Kumar, "Design, modeling, estimation and control for aerial grasping and manipulation," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2011, pp. 2668–2673.
- [18] B. Wang, Z. Ma, S. Lai, and L. Zhao, "Neural moving horizon estimation for robust flight control," *IEEE Transactions on Robotics*, vol. 40, pp. 639–659, 2024, open source: RCL-NUS/NeuroMHE.
- [19] M. Basseville and I. V. Nikiforov, *Detection of Abrupt Changes: Theory and Application*. Prentice-Hall, 1993.
- [20] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger, "Learning-based model predictive control: Toward safe learning in control," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 269–296, 2020.
- [21] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022.
- [22] M. Bangura and R. Mahony, "Nonlinear dynamic modeling for high performance control of a quadrotor," in *Australasian Conference on Robotics and Automation (ACRA)*, 2012.
- [23] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 3, 2004, pp. 2149–2154.
- [24] R. Verschuere, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [25] G. Frison and M. Diehl, "HPIPM: a high-performance quadratic programming framework for model predictive control," in *IFAC-PapersOnLine (21st IFAC World Congress)*, vol. 53, no. 2, 2020, pp. 6563–6569.